Experiment No.: 7

Date of Performance :
Date of Submission :

Aim: Data transfer using Bluetooth and analysis of performance.

Required Equipment / Software :

- Two Bluetooth-enabled smartphones, preferably Android devices.
- Built-in Bluetooth settings and a supported file-sharing/Bluetooth application.
- Measuring tape or meter scale.
- Suitable laboratory, corridor or open test area.
- A small test file such as a text file or image.

Theory : What is Bluetooth?
Bluetooth is a short-range wireless communication technology used to exchange data between electronic devices without using physical cables.
It operates primarily in the 2.4 GHz ISM (Industrial, Scientific and Medical) band.
Bluetooth is commonly used for:

- Wireless headphones
- Speakers
- Smartwatches
- Keyboards and mice
- Smartphones
- Fitness trackers
- IoT devices
- File and data exchange

Basic Bluetooth Communication
A typical Bluetooth connection involves two devices:



Before communication can take place, the devices generally need to:

1. Enable Bluetooth.
2. Discover each other.
3. Select the required device.
4. Perform pairing/authentication.
5. Establish a connection.
6. Exchange data.

Performance Parameters :
Bluetooth performance can be analyzed using several parameters.

- **Transfer Time**
  Transfer time is the time required to send a file from the sender to the receiver.
  It is measured in:**seconds (s)**

- **Data Transfer Rate**
  The effective data transfer rate can be calculated as:
  Transfer Rate= Data Size / Transfer Time
  If file size is expressed in megabytes:
  Rate (MB/s)= File Size (MB) / Time (s)
  For Mbps:
  Rate (Mbps)= File Size (MB)×8 / Time (s)

- **Example:**
  Suppose a **20 MB** file takes **10 seconds** to transfer.
  Rate=1020 =2 MB/s
  Converting to Mbps:
  2×8=16 Mbps
  Therefore:
  Effective transfer rate = 16 Mbps

Procedure :

Part A — Pairing

1. Switch ON Bluetooth on both smartphones.
2. Open Bluetooth settings.
3. Search for available devices.
4. Select the second smartphone.
5. Confirm the pairing request/passkey.
6. Verify that both devices are paired.

Part B — File Transfer

1. Select a test file on the sender smartphone.
2. Select **Share → Bluetooth** or the equivalent Bluetooth-sharing option.
3. Select the paired receiver.
4. Accept the transfer on the receiving device.
5. Start the stopwatch when the transfer begins.
6. Stop the stopwatch when the transfer completes.
7. Record the transfer time.
8. Verify that the received file can be opened successfully.

Observation Table :

Sr. No	File Type	File Size (MB)	Transfer Time (s)	Transfer Rate (Mbps)
1	**Text**	1	1.2	6.67
2	**Image**	5	3.5	11.43
3	**Audio**	10	7.0	11.43
4	**Video**	25	18.0	11.11
5	**Large Video**	50	37.0	10.81

Program :

```
import matplotlib.pyplot as plt
# Experimental data
file_size = [1, 5, 10, 25, 50] # MB
transfer_time = [1.2, 3.5, 7.0, 18.0, 37.0] # seconds

# Calculate transfer rate
transfer_rate = []

for size, time in zip(file_size, transfer_time):
 rate = (size * 8) / time
 transfer_rate.append(rate)

# Display results
print("File Size (MB) Time (s) Rate (Mbps)")

for size, time, rate in zip(file_size, transfer_time, transfer_rate):
 print(f"{size:10} {time:6.2f} {rate:8.2f}")

# Plot
plt.figure(figsize=(8, 5))

plt.plot(
 file_size,
```

```
 transfer_rate,
marker='o'
 )

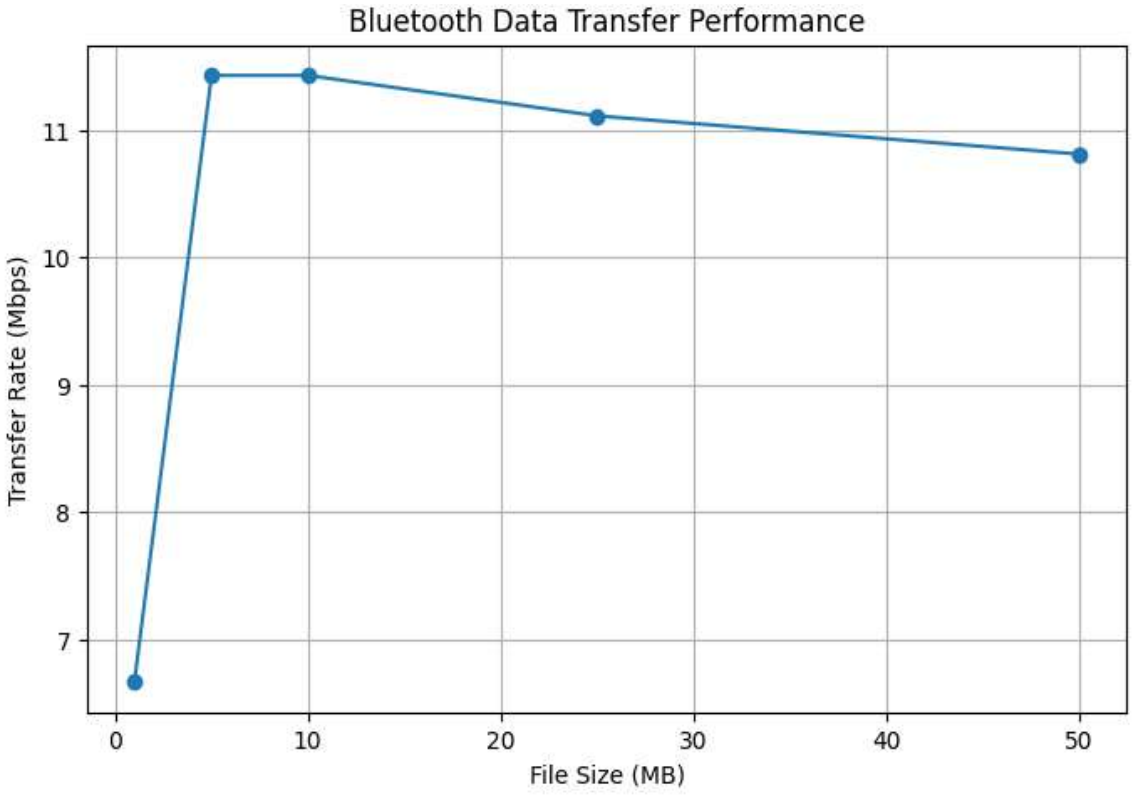
 plt.xlabel("File Size (MB)")
 plt.ylabel("Transfer Rate (Mbps)")
 plt.title("Bluetooth Data Transfer Performance")

 plt.grid(True)
 plt.show()
```

Output :

```
File Size (MB) Time (s) Rate (Mbps)
             1   1.20     6.67
             5   3.50    11.43
            10   7.00    11.43
            25  18.00    11.11
            50  37.00    10.81
```

Result :



Conclusion: The experiment demonstrates that Bluetooth can be used for wireless data transfer between smartphones without a physical connection. The measured transfer rate varies with file size and experimental conditions. For the given observations, the effective transfer rate is approximately 6.67–11.43 Mbps. Factors such as distance, interference, device capability, Bluetooth version, and environmental conditions can affect the actual performance.

Viva Voce Questions :

1. What factors affect Bluetooth performance?
2. What is Bluetooth Classic?
3. What security mechanisms are used in Bluetooth?
4. How does file size affect the experiment?

MARKS & SIGNATURE :

R1 (3 Marks)	R2 (5 Marks)	R3 (4Marks)	R4 (3 Marks)	Total (15 Marks)	Signature